

BOLZANO, ItalyWMO: **160200**Lat: **46.460N**Lon: **11.326E**Elev: **789**StdP: **14.28**Time Zone: **1.00 (EUC)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	19.1	21.4	0.2	5.7	35.5	6.1	7.7	36.8	17.3	42.6	14.1	44.7	1.9	150	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	92.9	70.5	89.7	69.4	87.3	68.5	72.6	88.3	71.2	86.1	70.0	83.9	8.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	105.1	78.0	66.3	99.8	76.3	65.0	95.3	75.1	36.6	88.2	35.4	86.8	34.4	84.0	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.2	12.1	14.1	96.4	2.0	4.3	12.7	99.5	11.5	102.0	10.4	104.4	9.0	107.5		
(5)			13.2	74.9	2.3	1.1	11.6	75.7	10.3	76.4	9.0	77.0	7.3	77.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.5	35.1	39.8	49.5	56.8	63.9	71.8	74.2	73.2	65.1	56.0	44.2	34.9		
	DBStd	15.05	4.76	5.26	6.01	5.90	5.44	5.85	4.85	5.42	5.10	6.12	6.16	5.54		
	HDD50	1519	462	287	82	11	0	0	0	0	0	18	191	468		
	HDD65	4373	927	705	481	250	87	13	3	5	60	284	625	933		
	CDD50	3511	0	2	65	216	432	654	749	720	453	204	16	0		
	CDD65	891	0	0	0	5	54	217	287	260	63	5	0	0		
	CDH74	8439	0	0	13	129	568	2111	2776	2373	450	19	0	0		
	CDH80	3197	0	0	0	18	143	843	1144	964	85	0	0	0		
(14) Wind	WSAvg	3.6	2.3	3.0	4.3	4.7	4.9	5.1	4.8	4.2	3.5	2.5	2.2	2.1		
(15) Precipitation	PrecAvg	28.5	1.1	1.1	1.9	1.9	3.3	2.9	3.7	3.9	2.9	2.5	2.2	1.1		
	PrecMax	42.3	4.8	3.6	4.5	3.4	7.6	5.7	9.4	10.0	10.2	7.8	6.6	3.5		
	PrecMin	16.5	0.0	0.0	0.0	0.0	1.0	0.5	0.7	0.9	0.1	0.0	0.0	0.0		
	PrecStd	6.3	1.0	1.0	1.3	1.0	1.6	1.3	1.9	2.2	2.7	2.3	1.8	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	63.4	75.4	82.4	88.1	95.0	95.6	97.4	86.7	76.6	64.0	53.9		
		MCWB	43.8	47.4	53.6	59.5	66.1	71.5	71.2	71.2	68.6	62.8	56.0	42.8		
	2%	DB	50.1	57.1	70.1	77.3	84.1	91.4	91.8	92.9	82.3	72.0	59.0	50.1		
		MCWB	39.6	44.2	51.9	58.7	64.4	69.9	70.2	70.6	66.5	60.4	51.9	41.5		
	5%	DB	46.6	53.6	66.2	73.6	80.3	88.2	89.5	88.2	78.9	69.5	55.8	46.6		
		MCWB	38.4	42.3	50.1	56.5	62.9	69.1	69.1	69.2	64.1	59.6	49.6	40.4		
	10%	DB	43.2	49.9	62.2	70.0	76.7	85.2	86.3	85.6	75.6	66.1	53.5	43.8		
		MCWB	36.3	40.5	48.8	55.2	60.7	67.7	68.4	68.5	62.9	57.6	48.8	39.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.8	48.8	55.2	61.9	69.0	74.5	74.1	73.8	70.5	64.9	57.4	47.3		
		MCDB	53.4	62.0	72.7	77.8	83.1	88.6	90.2	89.4	83.7	73.4	61.9	49.6		
	2%	WB	42.0	45.6	53.1	59.9	66.2	72.2	72.2	72.3	68.2	62.6	54.3	43.8		
		MCDB	46.9	53.1	66.6	74.6	80.5	87.6	87.7	88.2	78.1	69.5	57.1	46.6		
	5%	WB	39.6	43.7	51.4	58.0	64.2	70.6	70.9	70.8	66.4	60.8	52.2	41.7		
		MCDB	43.9	50.1	63.1	71.0	76.4	85.2	85.6	85.4	75.2	66.7	54.5	44.9		
	10%	WB	37.7	42.0	49.8	56.1	62.4	69.0	69.8	69.4	64.7	59.0	50.1	39.5		
		MCDB	41.7	47.5	59.8	67.2	73.5	82.4	83.6	82.5	73.1	64.3	52.3	43.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	19.5	22.5	22.0	22.1	23.1	23.3	22.7	20.7	17.9	16.4	16.6		
		MCDBR	20.8	25.5	30.9	28.7	27.9	27.3	27.4	27.4	24.5	21.3	19.7	19.9		
		MCWBR	13.0	14.4	15.6	13.1	11.7	10.3	9.7	9.5	11.0	11.2	12.6	12.8		
		MCDBR	18.9	21.3	26.4	26.2	24.5	25.3	24.4	24.2	21.1	18.2	14.6	16.6		
	5% WB	MCWBR	12.5	12.4	13.6	12.4	10.7	10.3	9.7	9.3	9.8	9.9	10.2	11.3		
(40) Clear-Sky Solar Irradiance	taub		0.297	0.332	0.393	0.457	0.472	0.488	0.478	0.468	0.431	0.398	0.342	0.300		
	taud		2.325	2.234	2.146	2.053	2.074	2.094	2.136	2.178	2.247	2.290	2.390	2.399		
	Ebn at Noon		251	262	260	253	254	250	251	248	246	235	232	237		
	Edh at Noon		24	32	41	49	50	49	47	43	37	31	23	20		
(44) All-Sky Solar Radiation	RadAvg		521	823	1220	1441	1570	1795	1818	1581	1228	820	502	431		
	RadStd		56	93	148	199	166	131	129	139	118	107	82	52		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (85 neighbors)		+0.65	N/A	N/A	+1.13	+0.65	+0.56	N/A	N/A	+119	+29

Nomenclature: See separate page